

Unit Tests :: CHEAT SHEET



About

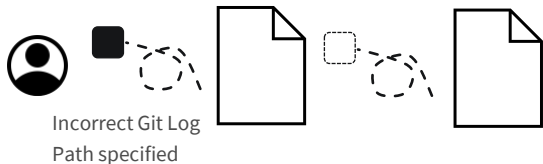
Kaiaulu has multiple measures in place to ensure that its codebase remains maintainable.

Try-catch statements and tests let functions fail gracefully and ensure functions work as intended.

Codecov and GitHub Actions run tests and other checks for every commit in a Pull Request, checking whether any tests fail, docs are missing, or other conditions are not met.

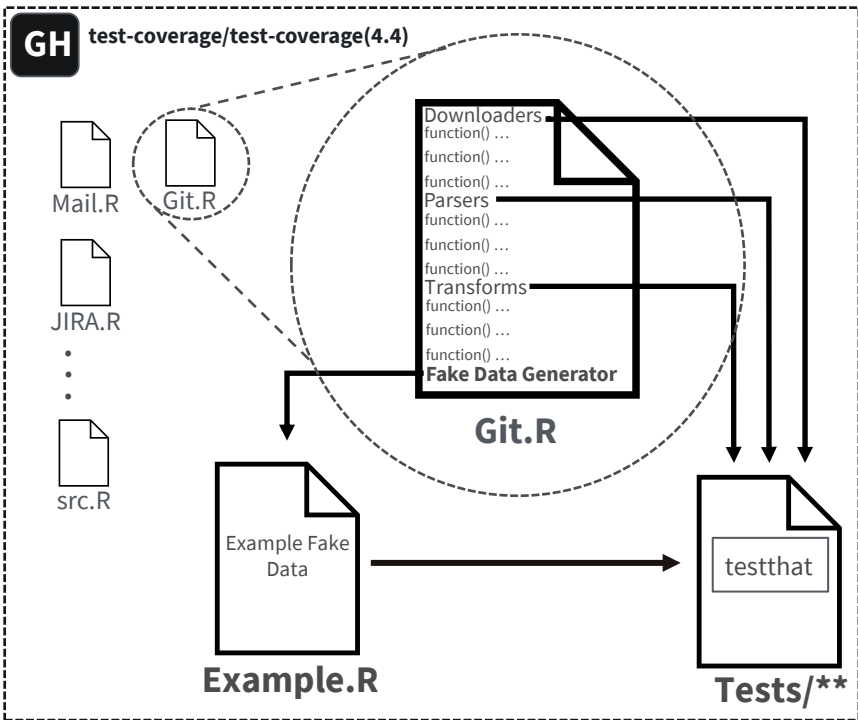
Failing Gracefully: Try Catch

Try catch statements allow errors that propagate through files to be caught gracefully.



A common example in Kaiulu is an invalid Git Log Path specified in a parser, which eventually throws an error in an .Rmd file.

Test Framework



The Fake Data Generator is used to construct examples, and those examples are used to test parsers for raw data.

Fake Data Generator

Fake Data Generation functions are at the bottom of each R/ file. They are called in Example.R to create the fake data cases.

Examples

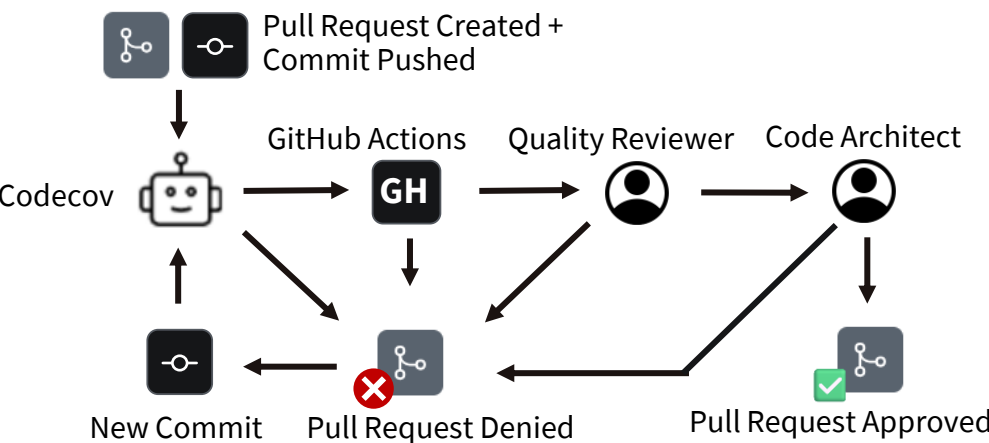
Kaiaulu's Example.R is used to create fake data examples. These mimic real raw data unexpected cases.

Tests

Tests for Kaiulu are made using R's testthat framework. Fake examples are used for parser tests, which parse fake data.

Code Review Workflow

The Kaiulu code review workflow follows a multiple step process.



Codecov

Codecov will deny a commit based on code test coverage.

GitHub Actions

GitHub Actions will deny a commit if any tests from the tests folder fail.

Quality Reviewer

The Quality Reviewer is a developer role that runs the committed code on their own machine, ensuring that users can replicate the capability or analysis.

Code Architect

The Code Architect is a user role that assesses if the committed code matches the conventions of other similar files in Kaiulu, such as function signatures, code location, and nomenclature.

Pull Request Workflow

In practice, the Code Review Workflow happens in the Pull Request. Codecov is represented by a bot comment on the pull request.

GitHub Actions (R-CMD-Check, test-coverage) respectively assess whether the docs are presented for the added API and whether the tests pass.

The Quality Reviewer and Code Architect then perform full code review.

Code	Issues	Pull Requests	Actions	Wiki	Security
Pull Request #330					
Username commented on Mar 15					
...					
Codecov commented on Mar 27					
✗ Patch coverage is...					
✓ Project coverage is...					
⚠ Report is...					
Changes Reviewed					
✓ 1 approval					
Quality Reviewer Approved these changes					
1 requested change					
Code Architect Requested changes					
Some checks were not successful					
1 failing, 1 successful checks					
✗ GH test-coverage/test-coverage(4.4)					
✓ GH R-CMD-check					
No conflicts with base branch					